



SILICA QUARRY
RECLAMATION PLAN
MARTIN MARIETTA CEMENT
WESTERN DIVISION
LYONS, COLORADO

Prepared By
Marvin L. Berg, P.E.
Quarry Superintendent
Martin Marietta Cement

77-361

RECEIVED

SEP 29 1977

MINED LAND RECLAMATION
Colo. Dept. of Natural Resources

MARTIN MARIETTA CEMENT

WESTERN DIVISION
CIVIL ENGINEERING DEPARTMENT
TULSA, OKLAHOMA 74104
TELEPHONE (918) 663-0030

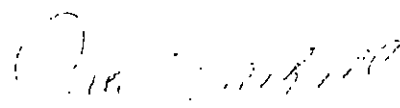
DALE MITCHELL
President

August 29, 1977

State of Colorado
Mined Land Reclamation Board
Department of Natural Resources
1313 Sherman Street
Room 723
Denver, CO 80203

Subject: Application for Mining &
Reclamation Permit

I, Dale Mitchell, as President of Martin Marietta Cement Western Division, have the authority to make sure the operator pursues and does the Application for Mining and Reclamation Permit as presented.



Dale Mitchell
President

The purpose of this report is to comply with the Mined Land Reclamation Act of 1976 and to make application for a permit to allow Martin Marietta Cement, Western Division, Lyons, to engage in a mining operation upon the land described in the application for the life of the mine. This application shall be considered under Statute 34-32-110 of House Bill No. 1065.

It is my intent to provide as much detailed information as possible pertinent to the mining operation and its effect on the ecology of the immediate area. The information and planning provided in this document has been prepared by Martin Marietta Cement and will constitute a working plan to be used for the remainder of the life of the plant.

The outline of this report follows the format in the Rules and Regulations of the Colorado Mined Land Reclamation Board, dated May 1977. I hope that the information is of sufficient detail and can be found and read easily.

Marvin L. Berg

Marvin L. Berg, P.E.
Quarry Superintendent
Martin Marietta Cement
Western Division
Lyons, Colorado

Reclamation Plan by:

Marvin L. Berg, P.E.

Member of Society of Mining Engineers of AIME

Member of National Society of Professional Engineers

Member of Professional Engineers of Colorado, Boulder Chapter

Information on Birds by:

Craig C. Williams

Vice President, Foothills Audubon Club

Federal and State Bird Bander

Submitted September 1, 1977

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STATE OF COLORADO
Mined Land Reclamation Board
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1313 Sherman Street, Room 723
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SEP 29 1977

Application for Mining and
Reclamation Permit, Pursuant
to C.R.S. 1973, 34-32-101 et seq.

MINED LAND RECLAMATION
State of Colorado Department of Natural Resources

File No. 77-361

1. Name of operations (Pit, mine, or site name)	Martin Marietta Silica Quarry
2. Name of applicant/operator (Name to be used on permit)	Martin Marietta Cement Western Division
3. Address and phone number of local offices	P.O. Box 529, Lyons, CO 80540 823-6685 534-4206 (Denver)
4. Address and phone number of general offices	P.O. Box 45586 Tulsa, OK 74145 (918) 663-0030
5. Name and phone number of person staff members should contact	G. L. Anderson 823-6685 P.O. Box 529 Lyons, CO 80540
6. Parent Corporation, if any (Include name & address)	Martin Marietta Corporation 6801 Rockledge Drive, Bethesda, MD 20034
7. Owner(s) of record of affected land (surface area) (If more than 2, list in Exhibit P)	Martin Marietta Corporation
8. Owner(s) of substance to be mined (If more than 2, list in Exhibit P)	Martin Marietta Corporation
9. Source of legal right to enter (Indicate what type of proof is included in Exhibit O - lease, deed, notarized letter, etc.)	Tax Receipts (1976)
10. Type of mine & substance to be mined (Specify underground, strip, open pit, etc. and substance(s) to be mined)	Open Pit Sandstone
11. General location & elevation (Miles & direction from nearest town & approximate site elevation)	1 mile S.E. Lyons, CO 5450 elevation
12. County	Boulder
13. Acreage of affected lands (Total surface acreage to be included in permit)	< 10 Acres
14. Application fee (See Rule 2 to determine amount)	\$125.00 (ck #1069)
15. Present land use (i.e., agricultural, industrial, etc.)	Agricultural
16. Surrounding land uses	Agricultural, Forest
Proposed future land use (Goal of reclamation)	Agricultural
Other mining permits & operations in Colorado (See Rule 2 for guidance)	Limestone Quarry - Lyons, Boulder Co. Limestone Quarry - Larimer Co.
The lands herein do not affect any of the lands described in Section 1.56(9), (Specify Affirmation)	Yes

OWNER OF RECORD OF AFFECTED LAND

OWNER OF SUBSTANCE TO BE MINED

Martin Marietta Corporation
6801 Rockledge Drive
Bethesda, Maryland 20034

OWNER OF RECORD OF AFFECTED LAND

OWNER OF SUBSTANCE TO BE MINED

Martin Marietta Corporation
6801 Rockledge Drive
Bethesda, Maryland 20034

(3) All lots which have been legally established prior to the enactment of this amendment shall remain as legal conforming lots.

6.4 MINIMUM LOT WIDTH . . . 140 ft.

6.5 MINIMUM LOT FRONTAGE . . . 40 ft.

(minimum front lot line)

6.6 MINIMUM FRONT YARD . . . 15 ft.

(minimum distance of any building from the front lot line, except as specified in Section XVIII)

6.7 MINIMUM SIDE YARDS . . . 25 ft.

(minimum distance of buildings from each side lot line)

6.8 MINIMUM REAR YARD

(minimum distance of any building from the rear lot line)

(1) dwellings and accessory buildings . . . 15 ft.;

(2) all other principal buildings . . . 25 ft.

6.9 MAXIMUM BUILDING HEIGHT . . . 35 ft.

SECTION VII

A - AGRICULTURAL DISTRICT

(AREAS WHERE CONSERVATION OF AGRICULTURAL RESOURCES IS OF MAJOR ECONOMIC VALUE)

7.1 USES PERMITTED

(1) One-family dwellings;

(2) Two-family dwellings;

(3) Agriculture, including nurseries, greenhouses, and the grazing of livestock and horses, provided the only products sold are those basically produced on the premises;

(4) Commercial feed yards, commercial poultry farms, fur farms, kennels, veterinary hospitals and riding academies, provided that all such uses are located at least 660 feet from schools, churches and

dwellings on other lots;

TAX ID	NAME	LEGAL	1ST HALF AMOUNT OF	REAL	PROPERTY TAXES PAID
AREA NO					
0870 057914	MARTIN-MARIETTA CORP	SW 1/4 20-3N-70 156 ACS	LAND VAL	410.00	PROPERTY TAX 16.58
	P.O. BOX 45586		IMP VAL	.00	
LEVY	1210 4TH NATL BLDG				
80.480	TULSA OKLA 74145		TOTAL VAL	410.00	
PAID BY	MARTIN-MARIETTA CORP		TOTAL TAX	33.16	TOTAL PAID 16.58 ** RECEIPT NBR A0302150997

THIS RECEIPT IS NULL AND VOID IF CHECK IS NOT HONORED BY BANK UPON WHICH DRAWN.

TAX ID	NAME	LEGAL	1ST HALF AMOUNT OF	REAL	PROPERTY TAXES PAID
AREA NO					
0800 054602	MARTIN-MARIETTA CORP	NE 1/4 NE 1/4 SE 1/4 SW 1/4	LAND VAL	14,010.00	PROPERTY TAX 607.32
	P.O. BOX 45586	NE 1/4 E OF HWY SE 1/4 NE 1/4	IMP VAL	8,220.00	
LEVY	1210 4TH NATL BLDG	LESS HWY S 1/2 S 1/2 SE 1/4 E			
51.630	TULSA OKLA 74145	DE HWY N 1/2 SE 1/4 E SE HWY	TOTAL VAL	22,230.00	
		LESS E 1/2 NE 1/4 SE 1/4 N 1/2			
		S 1/2 SE 1/4 E OF HWY LESS			
PAID BY	MARTIN-MARIETTA CORP	NE 1/4 SE 1/4 20-3N-70 148.44 AC	TOTAL TAX	1,014.64	TOTAL PAID 607.32 ** RECEIPT NBR A0302151007

THIS RECEIPT IS NULL AND VOID IF CHECK IS NOT HONORED BY BANK UPON WHICH DRAWN.

TAX ID	NAME	LEGAL	1ST HALF AMOUNT OF	PERSONAL	PROPERTY TAXES PAID
AREA NO					
0810 054735	MARTIN-MARIETTA CORP	LYONS COLO	PROP VAL	109,130.00	PROPERTY TAX 4,454.14
	WESTERN DIVISION		PRED VAL	.00	
LEVY	1111 SOUTH GOLDENADO BLVD				
81.630	DENVER COLO 80222		TOTAL VAL	109,130.00	
PAID BY	MARTIN-MARIETTA CORP		TOTAL TAX	8,908.28	TOTAL PAID 4,454.14 ** RECEIPT NBR A0302151017

THIS RECEIPT IS NULL AND VOID IF CHECK IS NOT HONORED BY BANK UPON WHICH DRAWN.

TAX ID	NAME	LEGAL	FULL AMOUNT OF	REAL	PROPERTY TAXES PAID
AREA NO					
0800 053385	MARTIN - MARIETTA CORP	E 1/2 NE 1/4 SE 1/4 E	LAND VAL	70.00	PROPERTY TAX 5.72
	P.O. BOX 45586	NE 1/4 SE 1/4 SE 1/4	IMP VAL	.00	
LEVY	1210 4TH NATL BLDG	20-3N-70			
81.630	TULSA OK 74145		TOTAL VAL	70.00	
PAID BY	MARTIN-MARIETTA CORP		TOTAL TAX	5.72	TOTAL PAID 5.72 ** RECEIPT NBR A0302151027

THIS RECEIPT IS NULL AND VOID IF CHECK IS NOT HONORED BY BANK UPON WHICH DRAWN.

BATCH M7

10
ACCOUNTS PAID

TOTAL PAID

293,200.15
226,000.15

CERTIFIED LISTING OF TAX RECEIPTS

Boulder County Treasurer

Boulder County Treasurer

Christine S. Stewart

By

1976 TAX RECEIPT

THOMAS J. MILBURN
BOULDER COUNTY TREASURER
P.O. BOX 471
BOULDER, COLORADO 80302

TAX DIST.	I.D. NUMBER
0870	067914

40692

THIS TAX RECEIPT IS FOR THE PAYMENT OF THE 2ND HALF OF YOUR PEAL PROP TAX

MILL LEVY	LAND VALUE	IMPROVEMENTS	PERSONAL PROP.	TOTAL VALUE	PRED. ANIMAL	TOTAL TAX
80.880	410.00			410.00		33.16

NAME AND ADDRESS OF TAXPAYER

MARTIN-MARIETTA CORP
P O BOX 45586
1210 4TH NATL BLDG
TULSA OKLA 74145

TAX DIST. AND
IDENTIFICATION
NUMBER

LEGAL DESCRIPTION

SW 1/4 20-3N-70 156 ACS

12047127

12047127

MAIL THIS RECEIPT TO THE ADDRESS BELOW

AMOUNT OF TAX PAID

16.58

INTEREST

ADVERTISING

TOTAL PAID

16.58

RECEIPT IS NULL AND VOID IF THE CHECK
OT HONORED BY THE BANK UPON WHICH
WN.

Thomas J. Milburn

TREASURER - BOULDER COUNTY

LEGAL DESCRIPTION

The Martin Marietta Silica property is described as the SW 1/4 Section 20 T3N R 70 W and includes 150.78 acres. The affected area of the quarry is described as 10 acres in the NE 1/4 SW 1/4 Section 20 T3N R 70 W of the 6th P.M.

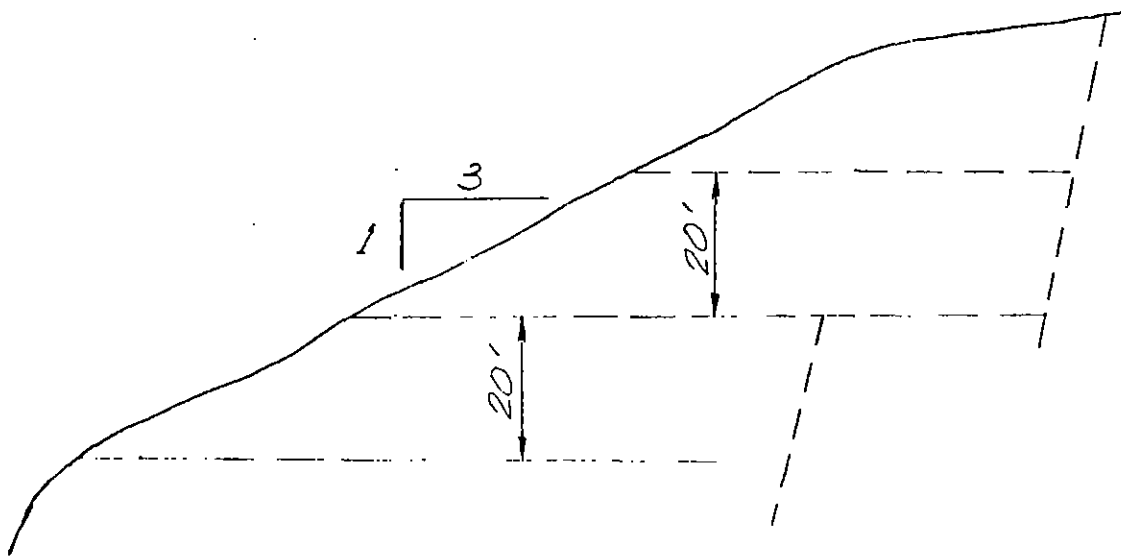
MINING PLAN

The Martin Marietta silica property is zoned A, agricultural. As stated in the Boulder County Zoning Resolution, Section VII, oil drilling, quarries, mining, sand and gravel operations and similar extractive land uses are permitted.

The mining plan for Martin Marietta silica Quarry, located one mile southeast of Lyons, Colorado is very straight forward. The quarry, operated by Loukenon Brothers of Lyons, Colorado, was purchased in 1974 from the Loukenon Brothers. At present 6.5 acres are opened up and will be expanded to a maximum of 10 acres. A very small portion of the operation is located on Loukenon property. This portion of the area is where a crusher and stockpile area are located.

The mine is located on a moderately steep side hill with only 0 to 8 inches of soil over the Dakota sandstone formation. This sparse soil is dozed off with a dozer and the sandstone is exposed. Mining is done in 20 foot horizontal benches proceeding west into the hillside. Since only 20,000 tons are mined each year, the advancement is very slow.

The following diagram describes the method of mining.



Typical Cross-Section

3" diameter holes are drilled on a 3 ft. by 3 ft. pattern. The sandstone is then blasted out of the hillside. The ore is transported to the crusher by a 3 c.y. loader and aided by a dozer. Once the material is crushed at the mine site, it is transported to the Martin Marietta Cement Plant one mile east.

RECLAMATION COSTS

Haul in material	\$2,500/Acre
Dozer (final grading)	250/Acre
Seed and fertilizer	250/Acre

Total Cost	\$3,000/Acre
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CONFIDENTIAL

WILDLIFE

The present fauna on the Martin Marietta Silica property has a rather low diversity with specific populations quite low. These observations have been made over the last seven to eight years while the quarry has been in operation. The observations, not recorded, were made by lay persons and in no way reflects a professional opinion.

A number of factors are possible contributions to the low populations of various wild animals:

- (a) Extensive farming and ranching has resulted in a landscape characterized by pastures and irrigated hayfields and croplands.
- (b) Extensive grazing by cattle and buffalo has reduced the cover for suitable habitat for many species.
- (c) The development and activity associated with the mining operations also affects the fauna of this region.
- (d) The entire open area is being reduced by housing and industrial projects. With a higher population density, the wildlife population will surely decrease.

The following is a list of animals and birds seen on Martin Marietta property. These animals and birds have been seen by Martin Marietta personnel since 1969.

Mammals

Cottontail Rabbits

These rabbits are very popular and can be seen almost daily.

Jackrabbits

These rabbits are also quite popular. Many of these rabbits are seen in the mine workings among the rocks and boulders.

Ground Squirrels

These squirrels are seen almost daily. It seems the mining operation has not had much affect on these animals.

Coyote

The coyote is common locally. Several have been sighted on the property. Some coyotes have been taken by predator control operations on adjacent properties in past years.

Raccoon

Raccoons have been seen along the ditches. A lot of signs have been seen which would indicate this animal is quite common in the area.

Skunks

Many skunks have been seen at various times. Skunks are quite predominant in farm and ranch areas.

Bobcat

Bobcats have been seen on the property. It is believed that the bobcat is a regular visitor to this area. The habitat is excellent for this species.

Mule Deer

Mule deer are quite common in this area. Although they tend to stay away from populated areas they will come down into the pastureland by the quarry.

Elk

These animals are quite popular in the winter months. Although they are not as popular as the mule deer, they can be seen from time to time.

Mountain Lion

None have been seen recently, but they used to be frequent visitors. It would not be unusual to see one.

Bear

Bear used to be quite common in past years. None have been sighted for some time now.

Birds

The present avian fauna on the Martin Marietta property appears to be moderately diverse and quite adequate for local and migrating birds. The property has coniferous and brushy ravine habitat.

The property is within an important north-south flyway for migrating birds as it borders the foothills and the plains. The property is in an ideal transition zone for birds as it lies in the foothills and weather changes will cause birds to move back and forth from the plains to the mountains.

Reclamation, presently ongoing on the property, will undoubtedly aid the bird population due to the planting of native and commercial seeds and grasses.

In addition to the bird-banding records of Mrs. Allegra Collister, we are also considering information from results of Christmas bird counts for the Longmont area (which includes the Martin Marietta property), as published in American Birds; results of sightings on the property by plant employees and local bird club groups; and assistance supplied from personal contact of members and use of records of the Foothills Audubon Club of Longmont.

Vultures

The turkey vulture has been observed in the area during the summer months, but is rather an infrequent visitor.

Accipiters

Goshawks are infrequent visitors to the area during the winter and early spring time.

Sharp-skinned hawks are very prominent in the area; probably nests in immediate area.

Harriers

Marsh hawks are common to the property but they primarily work open fields.

Buteos

Rough-legged hawks are common to the property during the winter season; improbable nester in the area.

Red-tailed hawk is very common and seen often by observers. Several instances of nesting with successful raising of young.

Swainson's hawk - several sightings but not very common to the area.

Eagles

Both the golden and the bald have occasionally been observed during the winter months, but, mostly, they are passing through the area and stay for very short periods of time.

Falcons

The sparrow hawk is the smallest of falcons and is very common to the property; it can be seen sitting on poles or telephone wires besides the plant roads or in the quarry areas.

Cuckoos

Yellow-billed cuckoo is an infrequent visitor to the property; it is found in the woods and brushy ravines during outbreaks of tent caterpillars.

Owls

Screech owls are infrequent in the heavy riparian cover.

Great horned owl - three plus pairs nesting on property, frequently seen.

Long-eared owl - Ideal habitat in brushy ravines on property.

Goatsuckers

Common nighthawk has been observed in flight at early evening hours, catching flying insects. Common to area of property.

Hummingbirds

Broad-tailed and rufous are frequent visitors to the property; others have not been observed, but during migration undoubtedly pass through property.

Woodpeckers

Red-shafted flicker is common and nesting in the wooded areas.

Lewis' woodpecker is uncommon to area but observed on occasion and could possibly nest on property in heavily wooded areas.

Yellow-bellied sapsucker is uncommon to the area the same as Lewis'; a little more infrequently observed on property.

Hairy and downy woodpecker - both species common and nesting on property in heavily wooded areas.

Flycatchers

Eastern and western kingbirds both observed on property. Western kingbird slightly more common on property than eastern species. The property has an abundant variety of flycatchers which are the willow, least, hammond's, dusky, western and western wood pewee.

Larks

Horned lark - large flocks (20-100) congregate on open areas during winter months. Very common.

Swallows

Barn, violet-green and rough-winged have been observed on property near ditches. Barn swallow would be the most common of the three and nests on the property.

Jays, Magpies and Crows

Stellar's jay have been observed in large numbers on the property. Blue jay and scrub jay have also been seen but in lesser numbers. All three prefer brushy or riparian habitat.

Black-billed magpie is common and conspicuous in lightly wooded and open areas; nests on the property.

Common crow is a frequent resident in wooded areas.

Chickadees

Both black-capped and mountain chickadee are common on property.

Nuthatches, Creepers

White-breasted, red breasted and pygmy nuthatches and brown creeper observed on property. All three prefer wooded areas, and the red-breasted probably nests on property.

Wrens

The house & canyon wren are fairly common in wooded areas, nest on property.

Mockingbirds and Thrashers

The mockingbird is an infrequent visitor to the property. Catbird, brown thrasher and sage thrasher are more common; prefer brushy habitat.

Thrushes, Solitaires and Bluebirds

The robin is a common and well known bird on the property; mostly inhabits brushy areas of property.

Townsend's solitaire - several sightings during nesting season and probably nests on or near the property in wooded areas.

Spotted-breasted thrushes - there are a limited number of hermit and Swainson's thrushes which inhabit the wooded understory; the veery is also common to property but prefers a wetter habitat than other thrushes.

Bluebirds - the western and mountain are common to the property but for only short periods of time as they migrate through this area to nesting areas in the foothills and mountains.

Gnatcatchers and Kinglets

Blue-gray gnatcatcher is common to the wooded areas of the property.

Ruby crowned kinglet is common to the property, as is the golden-crowned-kinglet.

Waxwings

Both the Bohemian and cedar waxwings are common to the property.

Shrikes and Starlings

Both the northern and loggerhead shrike inhabit the property with the loggerhead possibly a little more abundant. Can be seen perching on telephone wires in area.

Starlings are very abundant on property; inhabit lawns, open areas and farmland; nest on property.

Vireos

The three vireos common to the property are: solitary, red-eyed, and warbling; prefer shrubby and wooded areas; all three have been banded at Lykins Gulch.

Wood Warblers

Comprise the second largest percentage of birds observed and banded on the property. There is much ideal habitat for their nesting and subsistence contained on the property; warblers are almost entirely insectivorous; most nest within ten feet of the ground in brushy or wooded areas on the property; many nesting sites on the property at moderately wooded areas. Some of the mentioned warblers, of course, are more numerous than others, but, overall, the property supports a large and stable number of this species.

Black and White, Orange Crowned, Virginia's, Parula, Magnolia, Nashville, Black Throated Gray, Myrtle, Audubon, Hybrid Myrtle x Audubon, Bay-breasted, Blackpoll, Ovenbird, Northern Waterthrush, MacGillivray, Common Yellowthroat, Yellow-breasted Chat, Wilson's and American Redstart.

Weaver Finch

House sparrow are very numerous on property; open areas and farmland; nesting on property.

Blackbirds, Tanagers

Red-winged blackbirds are uncommon to the property but are seen occasionally feeding in open areas. Brewers blackbird common to farmlands on property. Common grackle western meadowlark and brown-headed cowbird also common to farmlands and open pasture lands. Western Tanager, Bullock's and Hybrid Baltimore x Bullock's Oriole are common to wooded and brushy areas and nesting on property.

Finches, Sparrows, Grosbeaks and Buntings

These comprise the largest family of North American birds. This family would occupy all property habitat. A large percentage of these mentioned will nest on the property; most prefer seeds although some will take berries, nuts from pinecones and insects.

Rose-breasted Grosbeak, Black-headed Grosbeak, Evening Grosbeak, Blue Grosbeak, Lazuli Bunting, Hybrid Lazuli x Indigo Bunting, Pine Siskin, American Goldfinch, House Finch, Lesser Goldfinch, Vesper Sparrow, Lark Sparrow, Clay-colored Sparrow, Tree Sparrow, Chipping Sparrow, Brewers Sparrow, Song Sparrow, Oregon Junco, Gray-headed Junco, White-crowned Sparrow and Lincoln's Sparrow.

Literature Cited

- (1) Reports of Christmas Bird Counts for Longmont Area.
- (2) Banding Records of Mrs. Allegra Collister.
- (3) Field trip Records of Foothills Audubon Club of Longmont.
- (4) Robbins, Bruun, Zim, Singer, 1966
Birds of North America

ENVIRONMENT

The presence of any rare or endangered species of plants or animals has not been revealed. Certain activities already in effect are significantly affecting the fauna and flora of this area. Some of these activities include (a) ranching, (b) predator control, (c) suburbanization.

There will, of course, be negative impacts upon the fauna and flora of the Martin Marietta property during mining operations. Recovery should be rapid and substantial. Post-reclamation conditions should be substantially improved over pre-mining conditions. At this time, it is difficult to say if the fauna will be superior to that presently existing. Due to rapid urbanization of the area along the front range, it is impossible to predict what the fauna will be like in 25 years.

In general, the dryland pasture will provide grazing for farm animals such as cattle and horses. Overall, the area will be very productive and will conform to the existing land use, flora, and fauna.

It is the intent of the reclamation plan to restore the land to as near the original land use. In many cases, the changing of contours will make the land more productive or desirable. The reclaimed land will be better overall and will make the environment very desirable for future generations.

WATER RESOURCES

The South Ledge Irrigation Ditch traverses the east boundry of the property. This is the only water resource located on this property. The drainage is generally from west to east coming off the foothills. There are no known aquifers on this property.

CO-RANGE-6
(File Code No. Range-18)
7/75
FOR RANGE - 18000000

U.S. Department of Agriculture
Soil Conservation Service

11/11/75 11/11/75
Conservationist
5 23 77
Date

RANGE CONDITION INVENTORY

RANGE SITE Shallow Foot-hill SOIL UNIT Colluvial Land

LAND UNIT _____ LOCATION OF INVENTORY Above Lookout
Shallow Foot-hills Hwy #66

VEGETATION

List of Plant Species (Grasses, forbs and shrubs)	Estimated per- cent composition by weight	Percent in po- tential plant community
<u>Erigeron sage</u>	<u>10</u>	<u>20</u>
<u>Prickly Pear</u>		<u>t</u>
<u>Yucca</u>		<u>1</u>
<u>Cheatgrass (downy Brome)</u>	<u>20</u>	<u>8</u>
<u>Mik Vetch</u>		<u>t</u>
<u>Curly Thistle</u>		<u>t</u>
<u>Wild rose</u>		<u>t</u>
<u>Snake weed</u>		<u>3</u>
<u>Ball cactus</u>		<u>2</u>
<u>Boxwood</u>		<u>5</u>
<u>Mustard</u>	<u>10</u>	<u>3</u>
<u>needle + thread</u>	<u>15</u>	<u>25</u>
<u>Blue gramma</u>	<u>15</u>	<u>10</u>
<u>Japanese Brome</u>	<u>15</u>	<u>10</u>
<u>Brome tail</u>	<u>5</u>	<u>8</u>
<u>Miscellaneous</u>		
<u>Current</u>	<u>10</u>	
Totals	<u>100</u>	<u>100</u>

Range Condition * Good Percent _____

Ground Cover: 40 %. Total Annual Yield 3.50 lbs./ac. air dry

Description of Growing Season (amount of precipitation and distribution)

Comments: Use, availability, and other significant factors _____

* Condition may be lowered one condition class from Excellent, Good or Fair when total annual yield is significantly (25%) below the site's potential.

GUIDE FOR RATING RANGE TREND

Considering the potential for this site, class of livestock, season of use, recreational use, fire, and other influences, rate the following by giving each item a rating of 4 to 0. Interpolate between indicators for ratings 3, 2, and 1.

Trend Indicators

REPRODUCTION:

Major species of the potential plant community are producing seedlings, plants are of mixed ages; there is evidence of active tillers, rhizomes, stolons (4)

Major species not reproducing (0) 2

PLANTS RESIDUE AND CURRENT UTILIZATION:

Litter produced by major species is abundant for the site and current utilization is moderate (4)

Litter is absent and utilization is severe (0) 3

COMPOSITION CHANGES:

Dominant species of the potential plant community are maintaining their place in the stand (4)

Dominant species are absent or noticeably decreasing in percentage, while minor species and species not native to the community are increasing (0) 3

PLANT VIGOR:

Major species of the potential plant community are strong, healthy, productive and well rooted (4)

Major species are showing noticeable die-off (0) 2

SOIL SURFACE FACTORS:

Accelerated soil erosion is not evident. Past erosion signs have healed. Water intake for the kind of soil is favorable.. (4)

Accelerated erosion is obvious. Soil stability is poor as seen by failure of erosion signs to be healing. Water intake for the kind of soil is unfavorable (0) 2

Ratings between 11 and 20 inclusive = Improving Trend 12

Ratings between 0 and 10 inclusive = Declining Trend _____

NOTES

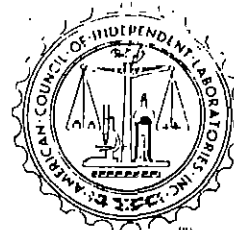
SOILS

The topsoil in this location is only 0 to 8 inches in thickness. This soil which is very rocky in nature sustains only the very tough dryland vegetation. Although no laboratory analysis was made at this particular site, an analysis from a quarry to the north is submitted. This analysis came from a location similar in nature and from the same foothills formation.

HAUSER LABORATORIES

5600 CENTRAL AVE. P.O. BOX G, BOULDER, COLORADO 80302 • PH. 303-443-4662

May 27, 1977
Test Report No. 77-370



CLIENT: Martin Marietta Cement Co.
P. O. Box 529
Lyons, Colorado 80540
Attention: Mr. M. L. Berg P.O.No. 10373

MATERIAL: Soil collected from Larimer County Quarry on 5-19-77.

TESTS: Soil tests as listed below.

RESULTS:	Test	Method	Results
	pH	USDA Handbook 60	7.7
	Conductivity, mmhos/cm	" " "	1.22
	Moisture Saturation %	" " "	46
	Sodium Absorption Ratio	" " "	.046
	Organic Matter, %	" " "	0.6
	Plant Available Phosphorus ppm as P ₂ O ₅	Bicarbonate Method	23
	Plant Available Potassium ppm as K ₂ O	USDA Handbook 60	120
	Particle Size Distribution	ASTM Sieves	
		greater than 2.0 mm	16.2%
		1.0	2.5
		.589	0.7
		.425	1.1
		.175	6.0
		.125	6.4
		.043	32.3
		less than .043	34.8

Tests Supervised By:

RECEIVED

JUN 2 1977

LYONS PLANT

Ronald D. Bruns

Ronald D. Bruns, Analytical Chemist

RESULTS FROM DESIGN AND RESEARCH

RECLAMATION PLAN

The primary objective for the reclamation plan for the Martin Marietta Silica quarry is to put the land back to as near the original land use as possible. The surrounding land is agricultural, made up of dryland pasture. We will restore the mining area to a dryland pasture environment.

The total area to be mined is <10 acres. Due to the lack of material, some overburden and rock will have to be hauled in from the Martin Marietta Cement quarry to the east. This material will be placed into the pit to obtain a maximum slope of 2 horizontal to 1 vertical. This field material will be covered with 2 to 4 inches of topsoil previously stripped off and stockpiled. The area will then be seeded with a dryland pasture mix and fertilized.

A dryland pasture mix presently being used at the Martin Marietta Cement Plant will be utilized. The following is a breakdown of the seed variety:

Dryland Mix

14.5	15	39% Winter Rye	6
1.5	15	13% Ranger Alfalfa	6
1.0	10	9% Smooth Broom	4
2.0	10	8% Intermediate Wheat Grass	4
1.0	10	7% Slender Wheat Grass	4
2.0	20	8% Nordan Crested Wheat Grass	2
1.0	10	6% Russian Wild Rye	4
2.0	10	10% Jose Tall Wheat Grass	4

25.0
25 pounds to 75 pounds of seed per acre will be planted. Ammonium Nitrate fertilizer will be applied at the time of seeding at a rate of 200 to 300 pounds per acre to facilitate growth. A portable sprinkler system will be used with water pumped from the South Ledge Ditch to initiate growth.

Reclamation will not be able to begin until the end of the anticipated life of this operation. Reclamation is planned to begin in the year 2025 and be complete in the year 2030.

To comply with Rule 6, Reclamation Performance Standards, the following statements will be according to the Rule 6 outline:

6.1 Grading

- a) All mined out areas will be backfilled. The land will not be backfilled with any toxic material. The material is compacted with the heavy equipment utilized in the mining operation.
- b) All grading is done on a slope of 2 horizontal to 1 vertical or less.
- c) The backfilling and grading is done as soon as practical after the mining process.
- d) No acid-forming or toxic-producing materials are mined at this site.
- e) There are no addits, shafts, or drill holes in the mining operation.
- f) All slopes are less than 2 horizontal to 1 vertical. There will be no proposed swimming areas and lakes.
- g) All the land will be reclaimed for pastureland.

6.2 Water

- a) There is no water discharge at this quarry.
- b) There will be no earth dams to impound water.

6.3 Wildlife

- a) Refer to exhibit on Wildlife Information.

6.4 Topsoiling

- a) Topsoil is stripped off and stockpiled for later use. Natural vegetation has come up and covered the piles.
- b) There are virtually no trees to speak of in the mining area.
- c) Stockpiles of topsoil are located on affected land.

6.4 Topsoiling (continued)

- d) Topsoil stockpiles are only rehandled once. They are not compacted except to the extent of when they are being built by heavy machinery.
- e) The slopes are graded to insure stability and prevent erosion.
- f) Topsoil used is the only material available.
- g) Topsoil is spread in 2" to 4" layers and fertilizer applied when seeding is done.
- h) There is no vegetative growth to be removed; consequently, no piles are made.

6.5 Revegetation

- a) Revegetation is accomplished by a grass mixture designed by Colorado State University.
- b) Forest planting is not part of the reclamation plan.
- c) All of the land will be restored to range, and all slopes are 2 horizontal to 1 vertical or less.
- d) The seed mixture being used in the reclamation plan has been very successful to date and has a high probability of success.
- e) The reclamation sites are prepared and graded by heavy equipment and then seeding and fertilizing are accomplished by a drill and farm tractor.
- f) Noxious weed will be sprayed when necessary.
- g) There are plenty of access roads to all locations of the property.